

The EOA Program, Part I: A Rank-Collapsed Deterministic Symbolic Representation From Convergent Recursive Sequences

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Abstract

We introduce a deterministic operator called beta. It turns the spoken names of letters into number sequences. When we apply it to the 26 letters of the English alphabet, it produces an infinite sequence for each letter, and we release the first 43 terms. This first part of the program, EOA-43, is a deterministic symbolic representation built from this recursive process.

After the first 38 terms, the ratio between one term and the next settles to a fixed number, λ . This ratio is the same for every letter, and it changes systematically if the phonetic encoding of the alphabet is changed (Part II [16]).

Because all 26 sequences share the same limiting ratio, they become nearly collinear as the terms grow. Any word vector built from them collapses to a low-dimensional subspace, effectively a single line. The 26 sequences group into 16 distinct sequence types and require 2.69 KB of storage. This small footprint follows from the representation's fixed low dimensionality.

This paper is the first in the series. Here we report the sequences, the grouping structure, and the algebraic consequences of convergence. Later papers will study how the limiting ratio depends on the encoding (Part II), the structure of the early terms (Part III), and how this representation behaves on real tasks (Part IV).

Keywords:

Alphabet engineering, deterministic representation, rank-1 representation, sequence collapse, limiting ratio, phonetic encoding, low-rank representation, training-free embeddings.

1. Introduction

Modern NLP systems segment text into tokens using methods such as Byte Pair Encoding (BPE) [2], which are learned from text data and stored in large, learned embedding tables. Standard embedding methods such as word2vec [4] and GloVe [17] rely on large training corpora and produce distributed representations that transfer across tasks. Alternative tokenization methods such as SentencePiece [24] address similar issues in multilingual and open-vocabulary settings, while tokenization-free encoders such as CANINE [12] and byte-level models such as ByT5 [13] bypass subword segmentation entirely. Character-level models [25] operate directly on raw text and provide a natural point of comparison for the EOA framework. The Engineering of Alphabets (EOA) program investigates whether a deterministic, symbolic decomposition of letter-name strings can yield useful word representations without training.

We asked a different question: can we build word vectors from nothing but the spoken names of the letters, by applying one deterministic rule?

The answer is yes, but the result is not a semantic embedding. It is a structural primitive. It carries almost no word meaning, but it has a rigid, repeatable structure.

The operator beta accepts a letter name (for example, "bee" for B, "aitch" for H) and applies a recursive transformation. After the first 38 terms, the ratio between one term and the next settles to a fixed number, $\lambda \approx 3.173$ (English modified #2). We call this the limiting constant ratio, or LCR.

The operator generates letter-specific transient structure in the early terms (terms 1 through 38) that is not captured by the limiting ratio alone. This transient structure is currently under investigation and may carry the representational information that distinguishes letters.

This paper documents what the sequences look like, how they group together, and what happens when you build word vectors from them. Because all sequences converge to the same ratio, the vectors collapse to a single effective dimension. We call this rank collapse. Part II [16] reports how the LCR shifts across different languages and encodings.

1.1 Scope and Positioning

This study does not claim that EOA replaces learned embeddings in general natural language processing tasks. Instead, the narrower conclusion proposed in this paper is that EOA-43 is a low-dimensional structural primitive. Its behavior on downstream tasks, including its profile under spelling errors, will be evaluated in Part IV. Its limitations are documented openly.

1.2 What EOA-43 Is Not

This representation is not a replacement for learned semantic embeddings such as word2vec [4], GloVe [17], or BERT [18]. It is not intended for tasks that require semantic understanding. It is not competing on downstream accuracy. Instead, it is a structural primitive whose algebraic structure yields extreme memory efficiency and rank collapse. Whether it also exhibits a near-flat noise profile will be evaluated in Part IV.

1.3 Contributions

1. A deterministic low-dimensional representation. EOA-43 vectors are computed from letter-name strings using 16 collinear sequence groups, requiring no training.
2. Memory efficiency with documented structure. The representation achieves a 94.4% storage reduction compared to GPT-2 embedding tables when cached, and eliminates static table storage under on-demand generation using the 16 distinct sequence groups (2.69 KB) (Section 4.2, Table 1). This reduction follows from the fixed 43-dimensional scheme. The specific contribution is the 16 distinct sequence-group structure: three letter groupings and 13 unique sequences derived from a single deterministic rule.

We note that the sequence-group structure is partially redundant with alphabetical index: letters sharing a sequence group are distinguished only by their alphabetical index in the rank-1 limit (Section 4.1). The 16 sequence-group amplitudes are therefore multipliers on an integer structure already present in the representation; this does not reduce the memory efficiency, but it clarifies that the compression is structural rather than algorithmic.

3. Rank-collapse consequence. The rank collapse is not an independent result. It follows from the shared limiting ratio, assuming the convergence holds in the released sequences. The full explanation is in Section 5.4 and Appendix A.1.

We also document the structural limitations that follow from this collapse. The representation has an effective rank of 1, and we report this limitation as part of the contribution rather than claiming broader expressiveness

1.4 History and Prior Circulation

A preliminary version of the sequence groupings and LCR constant was published on the Internet Archive in 2010 under the author's earlier transliteration, Bahaa M. Dergham [1]. The present paper does not include downstream classification tasks; those evaluations, with updated benchmarks and baselines, appear in Part IV.

1.5 Outline of the Series

Part I (this paper) presents the representation itself: its construction, its rank collapse, and its algebraic properties. Part II [16] presents the encoding-dependence of the underlying limiting ratio (LCR), showing that it varies systematically across encodings and is not a universal constant. Part III will examine the transient structure of the early terms. Part IV will map the empirical boundary of the representation on downstream tasks. Each part can be read independently, and together they build the full picture. The access and collaboration terms for this paper are described in Section 2. The broader program-level collaboration call appears in Part 0, Section 12.

2. Research Disclosure Program

This paper reports Phase 1 of a staged research disclosure. The empirical phenomena documented in this work (convergence to a fixed-point ratio, the 16 distinct sequence groups, rank-collapse behavior) can be independently verified using the 43-term sequences, which are available to qualified researchers under the terms described below.

Tier	Access	Agreement	Purpose
0	Published papers, sequence groupings, LCR values, benchmark code	None (public)	Priority and scientific disclosure
1	43-term sequences (English + extensible)	DUA, Data Use Agreement	Independent verification of empirical claims; research use only, no redistribution, citation required
2	Interactive playground (text to sequence)	ELA, Evaluation License Agreement + click-through Terms of Service	Interactive testing; hosted, rate-limited, no scraping, no ML training on outputs
3	Batch API (text file to CSV of sequences)	Research API License	Bulk evaluation; named-user, rate-limited, no data extraction
4	Oracle binary (black-box callable)	MTA, Material Transfer Agreement + Mutual NDA	Run on collaborator hardware; no source, no formula, anti-reverse-engineering clause, audit rights
5	Full operator beta and formula	TLA, Technology License Agreement + optional JDA, Joint Development Agreement	Commercial or funded collaboration; royalties, equity, or lump sum; co-inventorship on resulting patents

This graduated disclosure structure draws on practices from NIH (DUA), DeepMind AlphaFold (ELA/API), and pre-1997 RSA Laboratories (TLA). It is designed to enable independent verification of empirical claims (Tier 1) while protecting the generative mechanism (Tiers 4 and 5).

The operator is withheld under a staged disclosure policy while its theoretical scope is being mapped. The empirical claims (rank collapse, cross-encoding LCR variation, and memory footprint) can be checked at the output level using the released sequences and black-box oracle. A full specification will be released once the encoding-to-LCR mapping is theoretically characterized, which is the subject of active investigation.

For academic collaborators, access to the oracle (Tier 4) is available under a standard academic agreement that permits research use and publication of derived results with appropriate citation. The full commercial terms (Tiers 4 and 5) are not required for academic collaboration.

No agreement is required for Tier 0 materials. For access to Tiers 1 through 5, contact: bdarghamneurolabs@gmail.com

3. The Sequences

3.1 What the Operator Produces

A recursive operator beta generates the sequences used in this work. The operator itself is not disclosed. This paper reports only the released 43-term sequences and their observed properties. For each letter, beta produces an infinite sequence, and we release the first 43 terms. Terms 1 through 38 form the transient, and terms 39 through 43 form the converged tail.

The limiting ratio is:

$$\lim_{n \rightarrow \infty} \frac{s_{n+1}(c)}{s_n(c)} = \lambda$$

where $S_n(c)$ denotes the n -th term of the sequence for letter c .

For the English primary encoding, $\lambda \approx 3.306$. The sequences used in this paper come from the English modified #2 encoding, where $\lambda \approx 3.173$. Across all 26 released sequences from English modified #2, the limiting ratio is constant. Cross-encoding variation is reported in Part II [16].

The internal recurrence of beta is not disclosed in this paper. Section 2 describes the access tiers for the released sequences and for black-box access to beta. Whether the recurrence is linearly coupled has not been established, and no linearity test has been performed on the released sequences. If the recurrence were linearly coupled, the LCR would be independent of the input string, and random symbols or number words would be expected to converge to the same ratio. Preliminary control runs with random symbols and number words were performed, but their results are outside the scope of this paper and are reported in Part II [16].

In the converged tail, the 16 distinct sequence vectors are nearly scalar multiples of one another. Their early terms differ, and those differences are the subject of the transient analysis in Section 6. The released 43-term sequence vectors are provided in the JSON sequence file and are available to qualified researchers under the terms described in Section 2. The present empirical results do not rely on a formal proof of the internal recurrence.

3.2 Letter Groupings

Not every letter gets its own unique sequence. Some letters share identical 43-term outputs:

- Group 1: B, D, P, T, V, Z
- Group 2: F, L, M, N, S
- Group 3: I, R

The remaining 13 letters each have their own sequence. This gives 16 distinct sequence groups for the entire alphabet.

The fact that B, D, P, T, V, and Z share a sequence is itself an empirical finding. It suggests that these letters produce the same observable sequence and are distinguished in the representation only by their alphabetical index.

4. Building Word Vectors

4.1 Construction

To turn a word into a vector, we combine the sequences of its letters. Each letter is weighted by its position in the word and by its alphabetical index.

For a word of length k , let:

- $\text{seq}[c]$ be the 43-dimensional sequence of letter c ,
- j be the 1-based position of the letter in the word $(1, 2, \dots, k)$,
- $\text{idx}(c) \in \{1, \dots, 26\}$ be the alphabetical index of the letter ($a = 1, b = 2, \dots, z = 26$).

The EOA-43 vector is then:

$$\text{EOA}(\text{word}) = \frac{1}{k} \sum_{j=1}^k (\text{seq}[c_j] \times j \times \text{idx}(c_j))$$

This weighting makes the vector sensitive to both letter identity and letter order. Other weighting schemes are possible, but this paper follows this simple format because it directly encodes position and alphabetical identity without introducing learned parameters.

The result is a 43-dimensional vector. Because all 16 sequence groups converge to the same limiting ratio, their tail directions are collinear. In the released 43-term sequences, they are numerically collinear on the raw scale to within floating-point precision (see Section 5.3 for the standardized analysis). This means the 43 dimensions are not independent. The vector lies, for practical purposes, on a single line.

4.2 Memory Footprint

The numbers below compare two very different things: a learned semantic embedding table (GPT-2 BPE) and a fixed set of structural constants (EOA-43). Because EOA-43 stores no meaning, this is not a "semantic compression" claim. A random 43-dimensional projection with a fixed seed would need about the same tiny storage. Random projections [11] provide theoretical justification for low-dimensional embeddings, as established by the Johnson-Lindenstrauss lemma. This storage saving is not a property unique to the underlying operator.

Table 1. Memory comparison on the full GPT-2 vocabulary (50,257 tokens).

This comparison assumes an alphabetic word token for each vocabulary entry. Handling of subword tokens, punctuation, and special symbols is not defined in this paper and is not part of the claimed memory reduction.

Method	Dimension	Table Size	Reduction
GPT-2 (BPE)	768	147.24 MB	
EOA-43 (cached)	43	8.24 MB	94.4%
EOA-43 (on-demand)	43	0.0026 MB (2.69 KB, 16 sequence groups)	99.998%

All storage figures use binary units: 1 KB = 1024 bytes, 1 MB = 1024² bytes.

Where the number comes from. The released data contain 16 distinct sequences, each 43 numbers long. Stored as 4-byte floats: 16 times 43 times 4 equals 2,752 bytes.

The on-demand mode is the main memory advantage. You do not need to store vectors for every word. You store the 16 distinct sequence groups (2.69 KB) and build any word vector on the fly.

Table 2. Memory across vocabulary sizes (linear per-word scaling).

Vocab Size	BPE-768 (MB)	EOA Cached (MB)	EOA On-Demand (MB)	Reduction (Cached)
10,000	29.30	1.64	0.0027	94.4%
50,000	146.48	8.20	0.0027	94.4%
100,000	292.97	16.40	0.0027	94.4%
1,000,000	2929.69	164.03	0.0027	94.4%

No extra memory is needed as you add more words, for the same reason already stated: the representation is fixed and low-dimensional.

We therefore report 2.69 KB as the uncompressed reference storage for reproducibility.¹

¹ A smaller size may be possible. If we treat the 16 sequence groups as exactly collinear, one shared direction plus 16 amplitudes would take about 236 bytes in 4-byte floats. The 236 bytes do not include the 26-to-16 mapping table, and we have not measured the actual storage cost or checked how much accuracy would be lost.

5. Rank Collapse: A Mathematical Consequence

5.1 What "Low-Rank" Means Here

We measure effective rank as the number of principal components needed to capture 90% of the total variance. When the effective rank is low, most of the variance lies along a few directions, meaning the 43-dimensional vectors carry limited independent information.

5.2 What the Raw Vectors Show

Principal component analysis on the 16 distinct 43-term sequence vectors reveals that over 90% of the variance is captured by a single singular value. The effective rank is 1 (Figure A.1). For comparison, 16 random 43-dimensional vectors require many more components to capture the same variance. The Random-43 baseline is not a pure independent Gaussian; however, the EOA-43 rank collapse is a direct property of the shared limiting ratio, not a scaling artifact, and does not depend on that baseline.

Because the PCA analysis was performed on the 16 distinct sequence groups rather than on a large vocabulary of word vectors, the rank-1 result applies directly to these groups. The collapse of arbitrary word vectors follows by linear combination, not by a separate PCA over a word corpus.

5.3 Is the Rank-1 Result Just a Scaling Artifact?

The raw vectors are dominated by the tail of the sequence. With λ approximately 3.173 over 43 terms, the final term is about 10^{21} times larger than the first. This tail is identical for all letters because they all converge to the same ratio. So the raw rank-1 result is partly due to this exponential growth. But the more informative test is to look at standardized vectors, where each dimension is scaled to unit variance.

After standardizing the same 16 distinct 43-term sequence vectors, PC1 still captures over 90% of the variance, so the effective rank remains 1 (Figure A.1, standardized curve). The second component is floating-point numerical noise amplified by the scaling, but it does not affect the effective rank. This distinction clarifies that the rank collapse is not merely an artifact of scaling, but reflects the actual information content of the representation.

5.4 Why Does Rank Collapse Happen?

It is not accidental. If every distinct sequence vector converges to the same limiting ratio λ , then as the terms grow, each sequence vector is asymptotically proportional to the same geometric direction: $(1, \lambda, \lambda^2, \dots, \lambda^{42})$. Since all 16 distinct sequence vectors collapse to this single direction, any linear combination of them, any word vector, also collapses to the same direction. Rank 1 follows as a conditional consequence: if the sequences converge to a shared limiting ratio λ , then asymptotically each distinct sequence vector is proportional to the same geometric direction, and any linear combination collapses to rank 1. This derivation assumes the convergence property observed in the released sequences. It does not prove that convergence holds for the undisclosed recurrence. The derivation also depends on the empirical observation

that all 16 sequence groups share λ . The released data are available to qualified researchers under the terms described in Section 2.

Concretely, the word vector reduces, to first order, to a scalar coefficient proportional to:

$$\frac{1}{k} \sum_{j=1}^k j \cdot \text{idx}(c_j) \cdot a_{c_j}$$

where a_{c_j} is the scalar amplitude of the letter's sequence vector. The representation is effectively a position-weighted alphabetical index hash.

This first-order approximation captures the dominant geometric behavior of the representation. However, the operator's internal mechanism generates letter-specific transient structure in the early terms (terms 1 through 38) that is not captured by this scalar hash. Detrending the sequences by λ^n would isolate these transients. We hypothesize that any genuine representational information resides in this residual structure, and that the operator's value lies not in the limiting behavior but in these transients. This hypothesis is examined further in Section 6.2.

We verified this empirically: the mean cosine similarity between all 16 sequence vectors is 1.0000. Full details, including the derivation of the geometric direction and the supporting references, are in Appendix A.1.

6. Discussion

6.1 The LCR is Encoding-Dependent

The value $\lambda \approx 3.173$ reported here is specific to the English modified #2 encoding used for the released sequences. The English primary encoding yields $\lambda \approx 3.306$. In Part II [16] of this series, we show that the limiting ratio changes when the phonetic encoding changes. We tested seven encodings across four languages (English, Arabic, Hebrew, Greek), and observed LCR values ranging from 2.710 to 4.128. These observations indicate that lambda is not a universal constant; its value depends on the encoding. The consistency requirement is strict: the LCR only emerges when the encoding is applied uniformly across all letters. In our preliminary runs, changing a single letter while leaving the rest unchanged appeared to disrupt the global ratio. The full details of these perturbation tests will be reported in Part II [16].

6.2 The Transient as the Locus of Information

The limiting behavior of the sequences is geometric and therefore information-poor. If the sequences were purely geometric from term 1, the representation would reduce to a scalar hash. We hypothesize that the operator's representational information, if any, lies in the transient structure of terms 1 through 38. Isolating those transients by detrending with λ^n is a direction planned for Part III.

6.3 Relation to Prior Work

Some lightweight representations are training-free, such as character n-gram hashing and random projections. EOA-43 also requires no training, but it differs from these methods in its construction. It uses a fixed recursive decomposition of spoken letter names rather than hashing or random indexing.

FastText [3] is not training-free, but it shows how subword structure can improve word representations. EOA-43 is not a learned semantic embedding and is not intended to compete with FastText on downstream semantic tasks.

Feature hashing [9] and hash embeddings [10] offer memory-efficient representations through compact hashing schemes. EOA-43 also has a small memory footprint, but that small size comes from its low fixed dimensionality, not from hashing. No learned hash table or collision handling is involved.

Random projections [11] provide theoretical justification for low-dimensional embeddings, as established by the Johnson-Lindenstrauss lemma. EOA-43 shares their compactness, but its collapse does not come from random projection. It comes from the shared limiting ratio of the letter sequences.

Prior work has shown that neural NLP systems degrade sharply under synthetic and natural noise [7], and defenses via robust word recognition have been proposed [8]. EOA-43 is not a defense against noise. Whether its low-dimensional, semantically empty structure makes it less sensitive to noise is a hypothesis that Part IV will examine.

Symbolic dynamics [19] provides a natural framework for studying the convergence of such systems. Substitution dynamical systems provide a known class in which convergence ratios can arise from a dominant eigenvalue, as described by Perron-Frobenius theory [21]. EOA-43 may belong to a similar class, but its full recurrence is not yet disclosed, so this classification remains open. Matrix analysis [14] and matrix computations [15] underpin the numerical methods used in this work.

6.4 Concrete Use Cases

EOA-43 is best interpreted as a deterministic structural primitive for operational settings where static memory constraints are binding. Its combination of properties, a fixed 2.69 KB sequence set and deterministic on-demand generation, makes it relevant for memory-constrained environments. This storage saving is a function of low dimensionality, not a property unique to EOA-43's operator; a random 43-dimensional projection would yield the same footprint.

One illustrative scenario is on-device text assistance, where a compact representation could support lightweight suggestion or correction components without storing a large embedding table. The memory reduction is substantial: a standard BPE embedding table requires over 140 MB for a 50,000-word vocabulary, while EOA-43's on-demand sequence set requires only 2.69 KB, a reduction of roughly five orders of magnitude (Table 1).

Direct deployment-level validation on real-world tasks is left to Part IV of this series.

7. Limitations

This work has several documented limitations that define the scope of its contributions.

- **Semantic content:** EOA-43 is not designed to carry semantic information. Part IV will evaluate semantic similarity benchmarks such as WordSim-353. Based on the representation’s structure, we expect the correlation to be near zero.
- **Effective capacity:** The representation collapses to an effective rank of 1, limiting its expressive power.
- **Operator disclosure status:** The internal recurrence of the recursive operator is not disclosed. The mathematical properties reported here are empirical observations derived from the released sequence vectors and should not be interpreted as formal theorems about the operator.
- **Linear-recurrence hypothesis:** A preliminary test of the linear-recurrence hypothesis will be reported in Part II [16]. The result remains inconclusive. The derivation in Section 5 assumes the convergence property of the sequences; whether a linear system underlies this convergence is not confirmed.
- **Dataset breadth:** The downstream classification experiments planned for Part IV will use SST-2 as a single sentiment benchmark. Generalization to other tasks and domains will remain untested until those experiments are completed.
- **Baselines:** The downstream evaluation to be reported in Part IV uses character trigram hashing (Trigram-43) and random vectors (Random-43) as lightweight baselines. A direct comparison with FastText subword embeddings is deferred to future work.
- **Use-case validation:** The concrete use cases discussed in Section 6.4 are illustrative. Direct deployment-level evaluation on real-world tasks has not been performed.
- **Out-of-alphabet character handling:** The released 43-term sequences only cover the 26 English letters. Digits, punctuation, or special tokens are not part of this paper and will be addressed in Part IV.
- **Future directions:** First, detrending the 43-term sequences by λ^n (or equivalently, examining the residuals $\log s_n - n \log \lambda$) would isolate the letter-specific transients. We hypothesize that any letter-specific representational information is concentrated in these transients, but this has not been demonstrated. We leave their empirical characterization to Part III. Second, the reduced representation naturally yields a 1-D scalar baseline, the position-weighted alphabetical hash, which can be compared against the full 43-D EOA-43 vector in later work. Random-2 and Trigram-2 baselines could also test whether any observed noise robustness to be reported in Part IV is due to low effective dimensionality. These experiments are deferred to Part IV.

8. Conclusion

We have presented EOA-43, a deterministic word representation with a 16 distinct sequence groups and requiring a 2.69 KB memory footprint. Its defining algebraic property is an intrinsic rank collapse to 1 effective dimension (Section 5). With so little representational signal, the representation carries almost no semantic content, but it possesses a rigid, repeatable structure.

We do not recommend EOA-43 as a general-purpose embedding, but as a memory-efficient structural primitive. The empirical findings reported here are derived from the 43-term sequences released under the access terms in Section 2, and from the 16 distinct sequence groups. These materials are available for independent checking by qualified researchers. Because the internal recurrence of the operator remains undisclosed, the mathematical properties attributed to the operator should be regarded as empirical observations of its outputs until a full formal specification is published. We invite the research community to independently check, extend, and challenge these findings using the released 43-term sequence vectors and the code provided under the access terms in Section 2.

While this paper evaluates EOA-43 strictly as a structural primitive, its derivation from a recursive symbolic decomposition suggests that deeper structural properties may exist. Whether the 16 distinct sequence groups identified here reflect broader phonetic primitives, and whether the observed rank collapse can be harnessed for structural LLM compression or cross-lingual transfer, are open questions that we leave to future investigation. No empirical evidence for these directions is presented in this paper.

Part II [16] will report cross-encoding variation in λ across four languages and seven encodings. It will also address the theoretical question of which properties of an encoding determine its LCR. This remains an open problem.

Data Availability

The 43-term sequences for the English alphabet (`english_alphabet.json`) are not publicly released. The Colab notebook is configured to load them from a private source; without access to the sequence vectors, the download will fail and the notebook will not run. Access to the sequence vectors is available only to qualified researchers under the terms described in Section 2.

The code and Colab notebooks for verifying the structural results, excluding the sequence vectors, are available at the repository and in the provided Colab link under Appendix A.3. The code is provided under CC BY 4.0. The 43-term sequence vectors are provided under the access terms described in Section 2.

Conflict of Interest

The author has a proprietary interest in the undisclosed EOA operator and may pursue intellectual property or commercial applications related to it. No external funding was received.

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Appendix A: Supplementary Materials

A.1 Rank Collapse Verification

The rank collapse is a direct consequence of the shared limiting ratio. We verified this empirically using the released 43-term sequence vectors. At 64-bit floating-point precision, the 16 distinct sequence vectors, comprising three shared groups (B, D, P, T, V, Z; F, L, M, N, S; I, R) and 13 standalone letters, are numerically collinear on the raw scale: the mean pairwise cosine similarity is 1.0000 (min = 1.0000, max = 1.0000). Because the final term is roughly 10^{21} times larger than the first term, this observation alone cannot establish exact mathematical collinearity. Section 5.3 reports the standardized analysis, where the effective rank remains 1 but the collinearity is less exact.

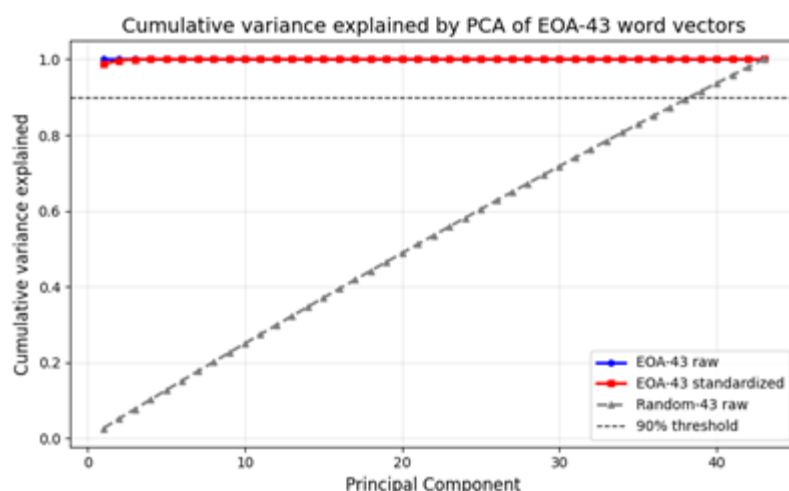


Figure A.1 Cumulative variance explained by principal components shows that the 16 distinct 43-term sequence vectors collapse to effective rank 1. For the raw vectors, PC1 captures more than 99% of the variance. After standardization, PC1 still captures over 90% of the variance. For comparison, 16 random 43-dimensional raw vectors require many more components to reach 90% variance. The 90% threshold is marked for reference.

A.2 Theoretical Background: Rank Collapse Derivation

The rank collapse is a mathematical consequence of the shared limiting ratio. If every distinct sequence vector converges to the same limiting ratio λ , then as the terms grow, each vector is asymptotically proportional to the same geometric direction: $(1, \lambda, \lambda^2, \dots, \lambda^{42})$. Since all 16 distinct sequence vectors collapse to this single direction, any linear combination of them, any word vector, also collapses to the same direction.

Symbolic dynamics [19] and Perron-Frobenius theory [21] provide theoretical frameworks that may eventually explain the convergence and rank-collapse properties reported here. They are not applied in the present paper and are noted as directions for the theoretical characterization planned in Part V. Matrix analysis [14] and matrix computations [15] are used here only as standard references for the numerical methods involved in PCA and similarity calculations.

A.3 Reproducibility Statement

The code and Colab notebook for checking the structural results are available here:

https://colab.research.google.com/drive/1WvLdr4qqDU_KY8H26DaYnR3N1U6CQIdV?usp=sharing

The notebook loads the 43-term sequence vectors from a private source. The sequences are available to qualified researchers under the access terms described in Section 2. They are not public.

Because the internal recurrence is not disclosed, the mathematical properties reported in this paper are empirical observations from the released sequences, not formal theorems about the operator. The operator remains undisclosed.